

GAEA-Net: Generating Activity-Enriched Abnormal ECGs via Adversarial Network

APPENDIX A: DATASETS

PTB-XL

PTB-XL is one of the largest publicly available ECG datasets, comprising nearly seven years of data collected between October 1989 and June 1996. The dataset includes 21,837 clinical 12-lead ECG recordings from 18,885 patients, each lasting 10 seconds. Each ECG signal was annotated by up to two cardiologists and categorized into five diagnostic classes: Normal ECG, Myocardial Infarction, ST/T Change, Conduction Disturbance, and Hypertrophy, with multiple classes potentially assigned to a single recording. Among all patients, 52% are male and 48% are female, with ages ranging from 0 to 95 years (median: 62, interquartile range: 22).

MuseECG

MuseECG consists of 10,646 patient ECGs, including 5,956 males and 4,690 females. Among these patients, 17% exhibited normal sinus rhythm, while 83% presented with at least one abnormality. The age groups with the highest prevalence were 51–60, 61–70, and 71–80 years, accounting for 19.82%, 24.38%, and 16.9% of the cohort, respectively. Each ECG recording is 10 seconds long, sampled at a frequency of 500 Hz. Each recording includes both rhythm labels and auxiliary diagnostic conditions.

MedalCare-XL

Unlike other datasets collected in clinical settings, MedalCare-XL was generated based on mechanistic cardiac electrophysiology models. As a result, no demographic information is available in this dataset. Two independent multi-scale models of atrial and ventricular electrophysiology were used to compute parameters of the P waves and QRS complexes, respectively. A total of 16,900 12-lead ECG signals were synthesized, including balanced representations of healthy individuals and seven classes of cardiac pathologies.

CinC 2017

CinC 2017, also known as the 2017 PhysioNet/Computing in Cardiology Challenge, focused on developing algorithms for the automatic classification of arrhythmias using single-lead ECG recordings. It comprises 8,528 single-lead ECG recordings with durations ranging from 9 to 61 seconds, sampled at 300 Hz. The recordings were acquired using the AliveCor portable device and labeled into normal sinus rhythm, atrial fibrillation (AF), and other abnormal rhythms.

PTB

The PTB Diagnostic ECG Database contains 549 15-lead ECG recordings (including the standard 12 leads and 3 Frank leads), acquired at a sampling rate of 1 kHz with 16-bit resolution (± 16.384 mV). These recordings were collected from 290 subjects aged 17 to 87 years, including 209 males and 81 females. The dataset includes various pathological conditions such as myocardial infarction, cardiomyopathy/heart failure, dysrhythmia, and myocardial hypertrophy.

APPENDIX B: BASELINE IMPLEMENTATION DETAILS

PGAN

PGAN introduces LSTM into GANs to generate plausible ECG signals, where the generator consists of a BiLSTM with 128 units. Under the instructions provided in its GitHub at <https://github.com/MikhailMurashov/ecgGAN>, we have rewritten the code using PyTorch for our experiments. We set the number of epochs to 100, and the batch size to 128 on all datasets. To address the issue of an overly strong discriminator during training, we adjust the learning rate of the generator to $2e-4$ and the learning rate of the discriminator to $1e-5$, training the discriminator once every five batches.

StyleGAN

StyleGAN3 is a state-of-the-art generative adversarial network (GAN) architecture primarily designed for generating high-quality images. We used the open-source code from the official repository: <https://github.com/NVlabs/stylegan3>. Although ECG signals are 1D, StyleGAN is typically applied to 2D images, so we adapted the approach by embedding the 1D ECG signals into 3-channel 2D images for training. Specifically, ECG time-series data are transformed into a 2D space using techniques such as GASF (Gramian Angular Summation Field), GADF (Gramian Angular Difference Field), and MTF (Markov Transition Field), which generate RGB-like 3-channel data. Finally, the original 1D ECG time-series are reconstructed using the diagonals of the GASF channel. For training, the setup used 3 GPUs with a batch size of 36, 2000k images (kimg), and a batch per GPU of 12.

SimGAN

SimGAN integrates a biological simulator into a Generative Adversarial Network to synthesize ECG signals, building upon the open-source implementation available at https://github.com/DreamStudioAI/sim_gan.git. The framework incorporates domain-specific knowledge from a heart simulator, represented by a system of Ordinary Differential Equations (ODEs), to guide the generative process. To adapt the method to our dataset, we modified the original euler loss function, which was based on a single heartbeat cycle, by dividing the generated ECG signal into 7 segments and calculating the loss for each segment individually.

CardioGAN

CardioGAN is an attentive generative adversarial network, featuring a U-Net generator with attention mechanisms and two specialized discriminators that evaluate both temporal and spectral signal characteristics. We utilize their open-source code available at <https://github.com/pritamqu/ppg2ecg-cardiogán>. Our implementation repurposes this architecture to translate between different ECG morphologies as categorized in the PTB-XL dataset, MuseECG Dataset and MedaCare-XL Dataset. Models were trained with a batch size of 128 for 40 epochs, using an initial learning rate of $1e-3$.

Diffusion-SSSM

Diffusion-SSSM integrates a diffusion model with a structured state space model (SSSM) to generate synthetic 12-lead electrocardiograms (ECGs) based on clinical data. This approach leverages the superior performance of diffusion models in data generation while incorporating the temporal dependencies inherent in ECG signals through the structured framework of the SSSM. To tailor the dataset to our specific experimental setup, we redefined the 71 original ECG statements as conditional categories. Additionally, we reduced the number of embedding dimensions accordingly to optimize model efficiency. The model was trained over 10,001 iterations, with a batch size of 4 and a learning rate set to $2e-4$.

ECG Twins

ECG Twins develops a vector-quantized feature separator that distinguishes between disease and normal segments of ECG signals under the guidance of diagnostic reports. To incorporate their methodology, we utilize their open-source code available at <https://github.com/huyjj/LAVQ-Editor>. Notably, since our datasets do not include pathology reports, we have omitted the text encoder component from the model architecture. During training, we set the number of epochs to 300, the learning rate to $1e-3$, and the batch size to 128 on all datasets.

APPENDIX C: ROC AND AUC RESULTS ON DIFFERENT DATASETS

PTB-XL

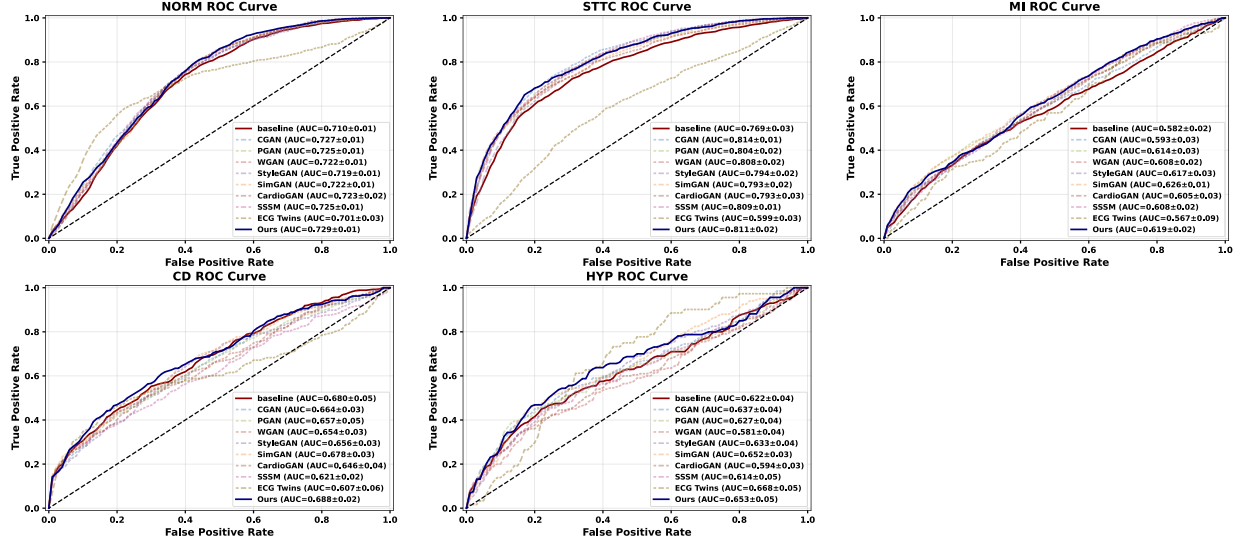


Fig. 1: ROC curves for the PTB-XL dataset

TABLE I: Class-wise AUC 95% confidence intervals of all methods on the PTB-XL dataset.

Method	Class1: NORM	Class2: STTC	Class3: MI	Class4: CD	Class5: HYP
Baseline	[0.703, 0.718]	[0.750, 0.788]	[0.573, 0.592]	[0.650, 0.709]	[0.600, 0.644]
CGAN	[0.721, 0.732]	[0.809, 0.820]	[0.578, 0.609]	[0.645, 0.682]	[0.610, 0.665]
PGAN	[0.722, 0.729]	[0.790, 0.819]	[0.596, 0.632]	[0.627, 0.687]	[0.601, 0.653]
WGAN	[0.716, 0.728]	[0.799, 0.818]	[0.594, 0.623]	[0.637, 0.670]	[0.558, 0.605]
StyleGAN	[0.714, 0.725]	[0.780, 0.807]	[0.601, 0.632]	[0.636, 0.676]	[0.606, 0.659]
SimGAN	[0.713, 0.731]	[0.778, 0.808]	[0.617, 0.636]	[0.663, 0.694]	[0.633, 0.672]
CardioGAN	[0.713, 0.733]	[0.777, 0.809]	[0.586, 0.624]	[0.621, 0.671]	[0.575, 0.612]
SSSM	[0.721, 0.729]	[0.801, 0.818]	[0.597, 0.620]	[0.607, 0.635]	[0.584, 0.644]
ECG Twins	[0.685, 0.718]	[0.583, 0.616]	[0.512, 0.622]	[0.573, 0.642]	[0.636, 0.699]
Ours	[0.723, 0.734]	[0.800, 0.823]	[0.604, 0.633]	[0.673, 0.704]	[0.616, 0.689]

MuseECG

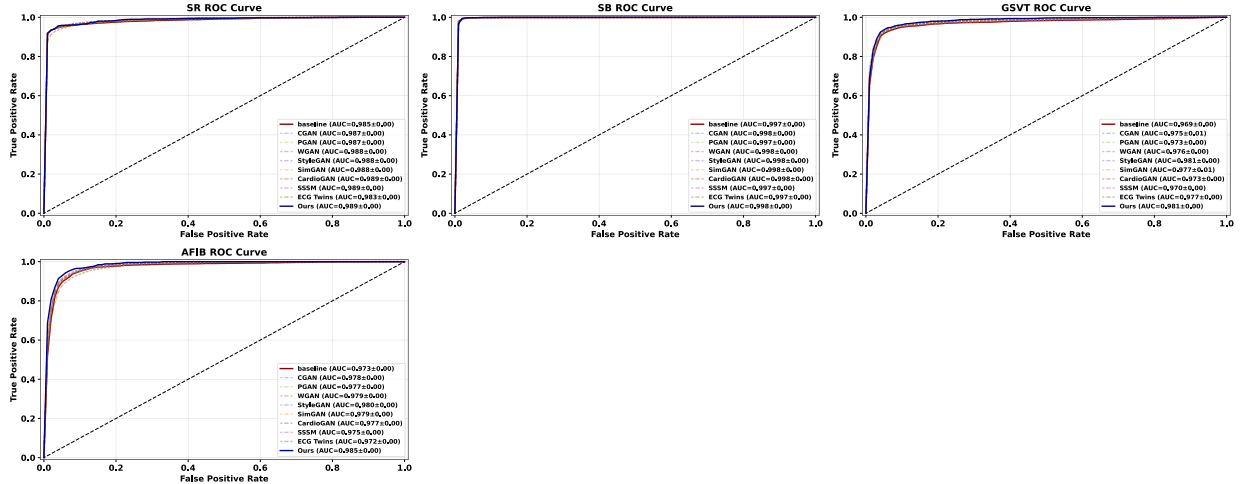


Fig. 2: ROC curves for the MuseECG dataset

TABLE II: Class-wise AUC 95% confidence intervals of all methods on the MuseECG dataset.

Method	Class1: SR	Class2: SB	Class3: GSVT	Class4: AFIB
Baseline	[0.984, 0.986]	[0.997, 0.997]	[0.966, 0.971]	[0.971, 0.974]
CGAN	[0.986, 0.988]	[0.997, 0.998]	[0.972, 0.978]	[0.976, 0.979]
PGAN	[0.986, 0.988]	[0.997, 0.998]	[0.971, 0.975]	[0.976, 0.978]
WGAN	[0.987, 0.989]	[0.997, 0.998]	[0.974, 0.979]	[0.978, 0.981]
StyleGAN	[0.987, 0.989]	[0.998, 0.998]	[0.979, 0.983]	[0.978, 0.982]
SimGAN	[0.986, 0.990]	[0.998, 0.998]	[0.973, 0.981]	[0.977, 0.982]
CardioGAN	[0.988, 0.990]	[0.997, 0.998]	[0.971, 0.976]	[0.976, 0.979]
SSSM	[0.988, 0.990]	[0.997, 0.997]	[0.968, 0.973]	[0.973, 0.977]
ECG Twins	[0.981, 0.985]	[0.996, 0.997]	[0.975, 0.980]	[0.970, 0.974]
Ours	[0.988, 0.990]	[0.998, 0.998]	[0.979, 0.984]	[0.985, 0.985]

MedalCare-XL

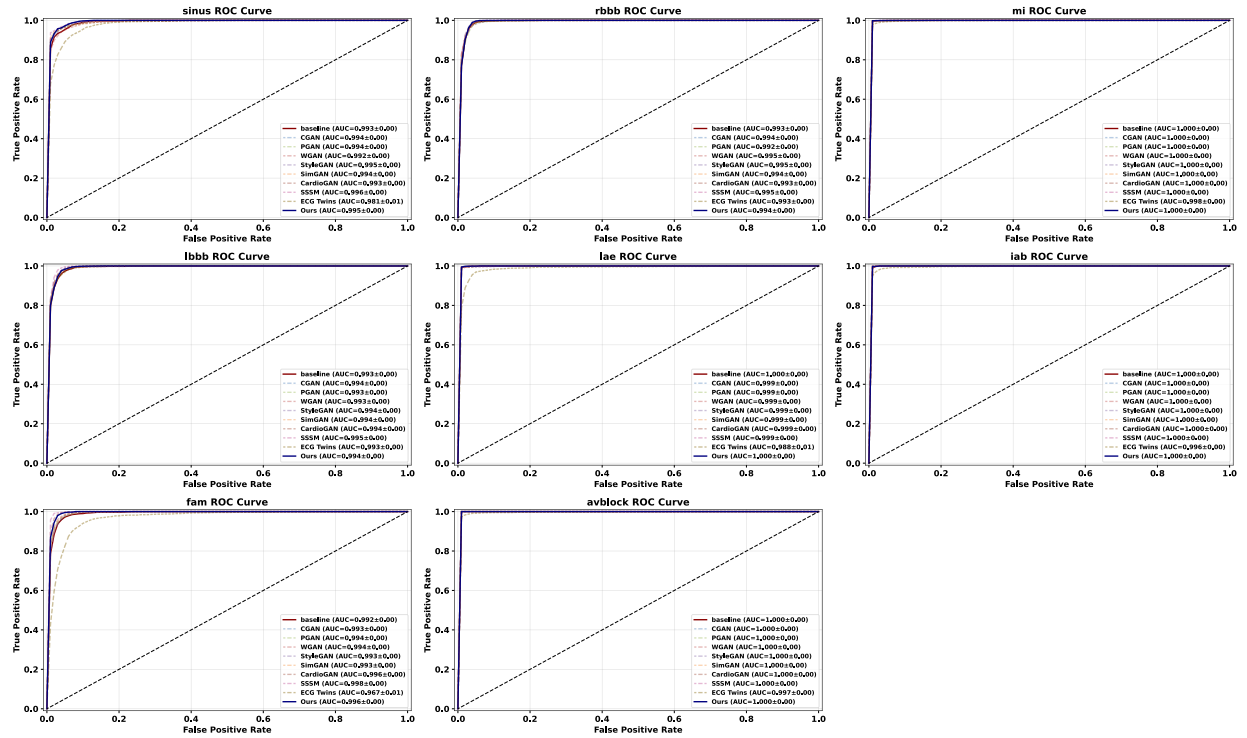


Fig. 3: ROC curves for the Medal-XL dataset

TABLE III: Class-wise AUC with 95% confidence intervals of all methods on the MedalCare-XL dataset.

Method	sinus	rbbb	mi	lbbb	lae	iab	fam	avblock
Baseline	[0.991, 0.994]	[0.992, 0.994]	[0.997, 1.000]	[0.991, 0.994]	[1.000, 1.000]	[0.999, 1.000]	[0.990, 0.993]	[1.000, 1.000]
CGAN	[0.992, 0.996]	[0.993, 0.995]	[1.000, 1.000]	[0.993, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.991, 0.996]	[1.000, 1.000]
PGAN	[0.993, 0.995]	[0.991, 0.994]	[1.000, 1.000]	[0.992, 0.994]	[0.999, 1.000]	[1.000, 1.000]	[0.991, 0.996]	[1.000, 1.000]
WGAN	[0.990, 0.994]	[0.994, 0.996]	[1.000, 1.000]	[0.991, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.992, 0.996]	[1.000, 1.000]
StyleGAN	[0.994, 0.996]	[0.994, 0.995]	[1.000, 1.000]	[0.994, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.991, 0.994]	[1.000, 1.000]
SimGAN	[0.993, 0.995]	[0.992, 0.995]	[1.000, 1.000]	[0.993, 0.994]	[0.999, 1.000]	[1.000, 1.000]	[0.992, 0.994]	[1.000, 1.000]
CardioGAN	[0.992, 0.994]	[0.992, 0.995]	[1.000, 1.000]	[0.993, 0.995]	[0.999, 1.000]	[1.000, 1.000]	[0.996, 0.997]	[1.000, 1.000]
SSSM	[0.995, 0.997]	[0.994, 0.995]	[1.000, 1.000]	[0.995, 0.996]	[0.999, 1.000]	[1.000, 1.000]	[0.998, 0.999]	[1.000, 1.000]
ECG Twins	[0.977, 0.985]	[0.992, 0.994]	[0.998, 0.999]	[0.993, 0.994]	[0.984, 0.992]	[0.994, 0.998]	[0.961, 0.974]	[0.996, 0.999]
Ours	[0.994, 0.996]	[0.993, 0.995]	[1.000, 1.000]	[0.993, 0.995]	[1.000, 1.000]	[1.000, 1.000]	[0.995, 0.997]	[1.000, 1.000]

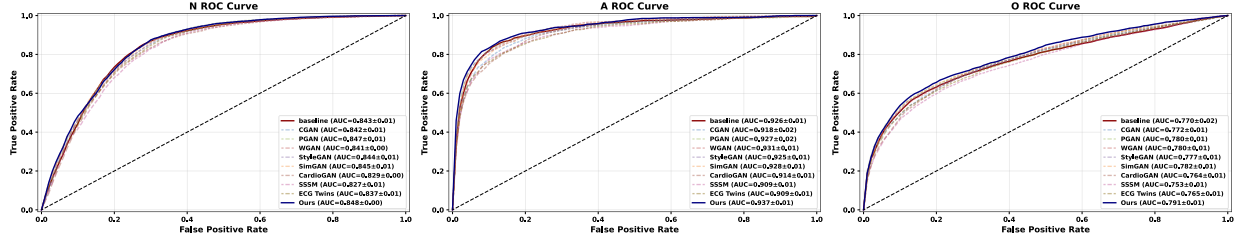


Fig. 4: ROC curves for the CinC 2017 dataset

TABLE IV: Class-wise AUC with 95% confidence intervals of all methods on the CinC 2017 dataset.

Method	Class1: N	Class2: A	Class3: O
Baseline	[0.839, 0.846]	[0.920, 0.932]	[0.759, 0.781]
CGAN	[0.837, 0.847]	[0.908, 0.927]	[0.764, 0.779]
PGAN	[0.841, 0.853]	[0.917, 0.936]	[0.776, 0.785]
WGAN	[0.838, 0.844]	[0.928, 0.934]	[0.775, 0.785]
StyleGAN	[0.839, 0.849]	[0.920, 0.931]	[0.772, 0.783]
SimGAN	[0.841, 0.849]	[0.923, 0.933]	[0.776, 0.788]
CardioGAN	[0.827, 0.831]	[0.908, 0.920]	[0.760, 0.769]
SSSM	[0.821, 0.834]	[0.901, 0.916]	[0.747, 0.759]
ECG Twins	[0.832, 0.842]	[0.900, 0.918]	[0.759, 0.770]
Ours	[0.846, 0.851]	[0.934, 0.941]	[0.786, 0.796]

PTB

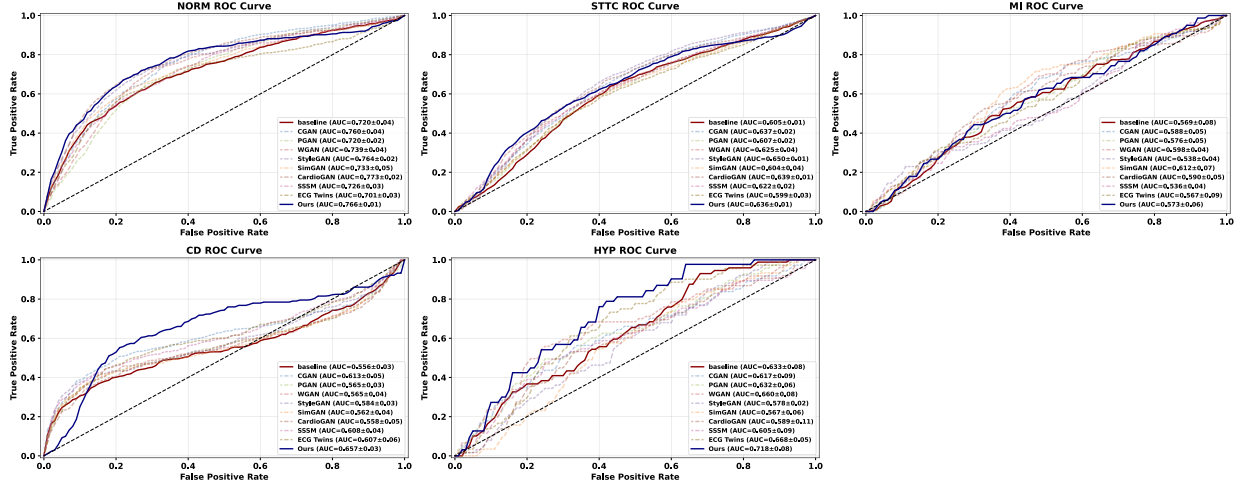


Fig. 5: ROC curves for the PTB dataset

TABLE V: Class-wise AUC with 95% confidence intervals of all methods on the PTB dataset.

Method	Class1: NORM	Class2: STTC	Class3: MI	Class4: CD	Class5: HYP
Baseline	[0.696, 0.744]	[0.598, 0.613]	[0.517, 0.621]	[0.536, 0.577]	[0.586, 0.680]
CGAN	[0.736, 0.785]	[0.626, 0.648]	[0.559, 0.617]	[0.580, 0.646]	[0.558, 0.676]
PGAN	[0.705, 0.736]	[0.596, 0.618]	[0.543, 0.609]	[0.545, 0.585]	[0.593, 0.671]
WGAN	[0.714, 0.764]	[0.600, 0.651]	[0.572, 0.624]	[0.540, 0.590]	[0.609, 0.710]
StyleGAN	[0.752, 0.776]	[0.642, 0.658]	[0.516, 0.560]	[0.568, 0.601]	[0.563, 0.593]
SimGAN	[0.702, 0.764]	[0.579, 0.628]	[0.566, 0.657]	[0.537, 0.561]	[0.530, 0.604]
CardioGAN	[0.758, 0.787]	[0.630, 0.648]	[0.562, 0.619]	[0.529, 0.587]	[0.520, 0.657]
SSSM	[0.705, 0.748]	[0.609, 0.636]	[0.511, 0.561]	[0.583, 0.634]	[0.551, 0.660]
ECG Twins	[0.685, 0.718]	[0.583, 0.616]	[0.512, 0.622]	[0.573, 0.642]	[0.636, 0.699]
Ours	[0.759, 0.773]	[0.631, 0.641]	[0.534, 0.613]	[0.640, 0.674]	[0.668, 0.768]

APPENDIX D: VISUALIZATION OF FEATURE SEPARATION

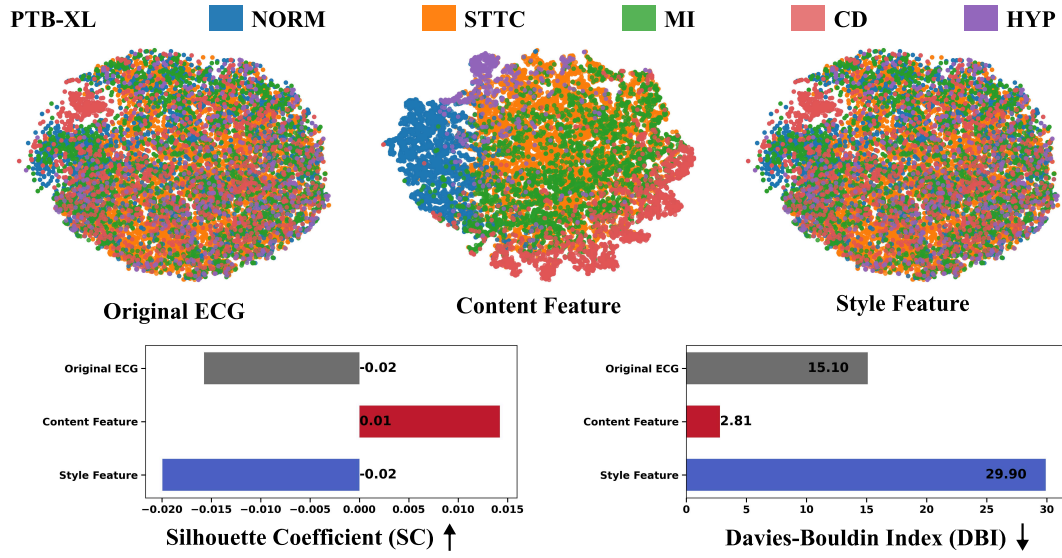


Fig. 6: Visualization of feature separation on the PTB-XL dataset

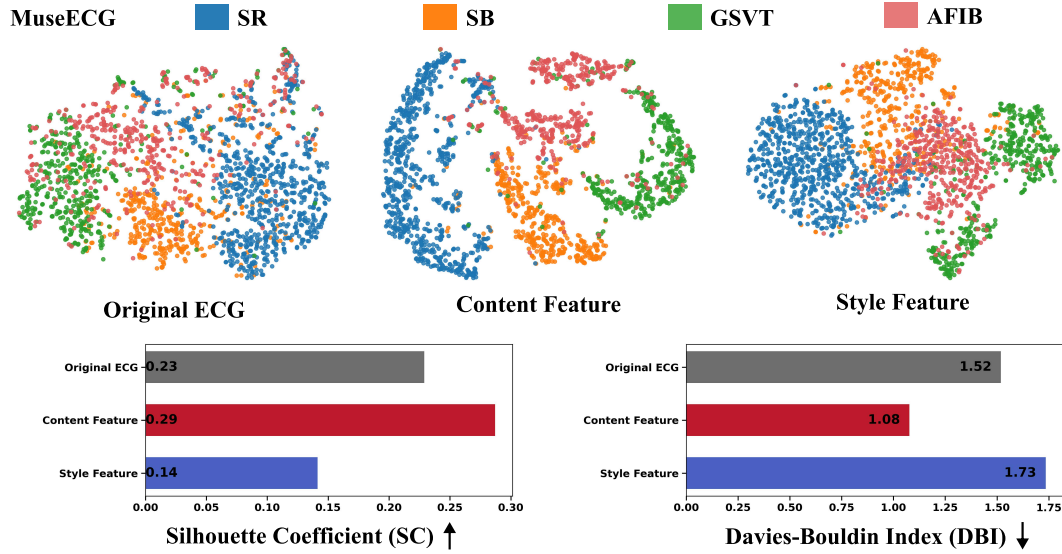


Fig. 7: Visualization of feature separation on the MuseECG dataset